

Joseph Kerry

West Bridgford, Nottingham • [GitHub](#) • [Portfolio](#)

Education

Master Of Science in Software Engineering at NOTTINGHAM TRENT UNIVERSITY

Overall grade: Low Distinction

Major Project thesis: Assessing the Galea Headset on a Benchmark of Deep Learning Architectures and Pre-processing Pipelines for EEG Motor-Imagery Task Classification

Employment

CoreFlow Operations Subcontractor (Feb 2026 – May 2026)

Part-Time Systems Developer/ Administrator

- Consulting documentation and requirements for SMEs.
- Developing systems from a user perspective to ensure usability and functionality.
- Partial development of Nottingham Forest Supporters Trust website in accordance with requirements.

Notable Projects

Assessing the Galea Headset on a Benchmark of Deep Learning Architectures and Pre-processing Pipelines for EEG Motor-Imagery Task Classification

- Developed a reproducible benchmark with a novel deep learning architecture that matched published architecture accuracy and provided unique findings related to the Galea headset.
- Hands-on experience with OpenBCI's Galea headset, used as an EEG cap for signal acquisition to determine external validity

Distributed Cloud Load Balancer

- Built a Java based distributed cloud storage simulator across four dockerised server nodes, implemented client-side AES encryption, file chunking and load balanced distribution.
- Designed a dual database architecture with an MQTT based host manager supporting dynamic scaling.

Convolutional Vision Transformer – Temporal Convolutional Network Hybrid Model

- Implemented a CVT-TCN using PyTorch for motor imagery classification (not for major project) on the EEGMMIDB dataset, converting raw time trials into time-frequency representations.
- Achieved a subject independent accuracy on 4 class motor imagery, achieving similar results to EEGNet with an end-to-end pipeline.

Machine Learning Library in C

- Implemented classical ML and DL algorithms in C, including supervised learning, clustering and neural network approaches
- Built on C11, including custom arena memory allocation, dense matrix and vector operations, Makefile unit test suite.

Skills

Languages: Python, C, Java

ML/DL: PyTorch, NumPy, MNE-Python, PyTest, SciPy, Jupyter, Scikit-learn

Databases: SQL, SQLite, PostgreSQL

Other technical skills: Docker, Git, Linux, YAML, JSON, C#, Java